Linux administration tips and tricks

Master the command line

- Get comfortable with the Linux command line.
- The command line your Swiss Army knife for DevOps tasks. Learn basic commands like ls, cd, cp, mv, cat, echo and more.

File Permissions

- Understand file permissions (chmod), ownership (chown), and groups.

  Managing access control is crucial in a DevOps role.

SSH Key Pairs

- Generate SSH key pairs for secure server access. Use ssh-agent to manage keys and never store private keys in plaintext

Firewall Rules

- Learn iptables or use firewalld to control network traffic.

  Properly configuring firewalls is vital for server security.

Take Regular Backups

- Set up automated backups using tools like rsync or tar.
- Regular backups can save your data in emergencies.

Scripting Skills

#!/bin/bash

- Become proficient in Bash or Python scripting to automate repetitive tasks.

  This is the heart of DevOps automation.

Package Management

- Know your package manager (apt, yum, dnf).

  Keep software up to date and resolve dependencies efficiently.

Network Troubleshooting

- Learn iptables or use firewalld to control network traffic.

  Properly configuring firewalls is vital for server security.

Process Management

- Use commands like ps, top, and systemctl to monitor and manage processes.
- Keep an eye on server performance.

Disk Management:

- Master commands like df and du to manage disk space.
- Use LVM for flexible storage management.

Version Control

- Get comfortable with Git for tracking changes to your code and configurations.
- Git is essential for collaboration.

Containerization

- Explore Docker and container orchestration tools like Kubernetes.

  Containers simplify deployment and scaling.

Configuration Management

Learn tools like Ansible, Puppet, or Chef to automate server configuration and ensure consistency.

Cloud Services

- Familiarize yourself with cloud providers like AWS, Azure, or GCP.
- Infrastructure as Code (IAC) is a game-changer.

Testing & CI/CD

- Integrate testing into your DevOps pipelines. Use Jenkins, Travis CI, or GitLab CI for continuous integration and delivery.

Monitoring & Logging

- Implement monitoring with tools like Prometheus, Grafana, or Nagios. Centralized logging (ELK stack) helps troubleshoot issues.

Infrastructure as Code

- Write infrastructure code with Terraform or CloudFormation to automate provisioning and scaling of resources.

Automation & Orchestration

- Use tools like Kubernetes, Docker Swarm, or Apache Mesos for container orchestration. Automation streamlines tasks.

Certifications

- Consider certifications like AWS Certified DevOps Engineer, Certified Kubernetes Administrator (CKA), or Red Hat Certified Engineer (RHCE) to validate your skills.

Continuous Learning

- Stay updated with the ever-evolving DevOps landscape. Attend conferences, read blogs, and keep experimenting to excel in your career.